ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Mercury, Venus, Earth, Mars	Order.
2	**(a)** complete one revolution around the Sun	Orbital period.
3	**(d)** it rotates once in the time it orbits the Earth	Synchronised.
4	**(d)** Venus: hottest planet	Right.
5	**(d)** rotate once on its axis	Rotation period.
6	**(d)** 29.5 days	Phase cycle.
7	**(c)** its temperature is suitable	Habitable zone.
8	**(b)** Neptune	Farthest from the Sun.
9	**(d)** 15 January	About two weeks later.
10	**(a)** sound cannot travel	No air.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Day is one rotation (1); Year is one revolution (1).	1+1
12	They are in free fall around Earth (1); Gravity still acts (1).	1+1
13	They are closer (1); They look like discs (1).	1+1
14	Rotation time equals revolution time (1); So the same side faces Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Earth has air and water (1); Moon has none (1); Moon has extreme temperatures (1).	1+1+1
16	It orbits Earth (1); It was not made by humans (1); Artificial ones are man-made (1).	1+1+1
17	Collects light (1); Focuses it (1); Magnifies image (1).	1+1+1
	OR: Moon orbits Earth in about a month (1); Lit part seen changes (1); Cycle repeats (1).	1+1+1
18	Moon's gravity is weaker (1); About one sixth of Earth's (1); Mass does not change (1).	1+1+1
	OR: Jupiter: largest, Great Red Spot (1); Saturn: rings (1); Both gaseous (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Locate Saptarishi (1); Extend line from the bowl (1); Pole star gives north (1); New moon starts month (1); Full moon near mid-month (1).	1+1+1+1+1
	OR: Water in liquid form (1); Oxygen (1); Right temperature (1); Protection by atmosphere (1); Distance from the Sun (1).	1+1+1+1+1
20	Earth: life, water (1); Moon: no air (1); Mars: thin air, cold (1); Venus: hot (1); Jupiter: gas, no surface (1).	1+1+1+1+1
	OR: Navigation (1); Calendars (1); Farming seasons (1); Stories (1); Pole star (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that the Moon makes its own light; it only **reflects** sunlight.
- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
- Confusing a star with a planet: the Sun is a star and the Earth is a planet.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Mercury	Nearest to the Sun.
2	**(c)** 8 minutes	Roughly 8 minutes.
3	**(a)** meteorite impacts	Impacts leave craters.
4	**(c)** Jupiter and Saturn	Large, gaseous.
5	**(c)** light year	Very large distances.
6	**(b)** Earth rotates from west to east	Relative motion.
7	**(c)** Mercury, Venus, Earth, Mars	Terrestrial.
8	**(d)** Pole star	It is a distant star.
9	**(b)** new moon	Dark side faces Earth.
10	**(b)** in their orbits	Pulls them.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Our galaxy (1); Contains billions of stars (1).	1+1
12	It collects more light (1); It magnifies (1).	1+1
13	It is nearest (1); Others are very far away (1).	1+1
14	Iron oxide on surface (1); Rust-coloured soil (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New moon: Moon between Sun and Earth (1); Full moon: Earth between (1); Lit face turns to or away from us (1).	1+1+1
16	Know about the Universe (1); Technology (1); Resources and safety (1).	1+1+1
17	It reflects sunlight (1); It has no light of its own (1); The Sun lights up half of it (1).	1+1+1
	OR: Crop monitoring (1); Cyclone tracking (1); Flood warnings (1).	1+1+1
18	Light and heat to planets (1); Powers weather (1); Plants use it (1).	1+1+1
	OR: Telescopes (1); Space probes (1); Rovers and orbiters (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Torch as the Sun (1); Globe rotates (1); Day on the lit side (1); Ball as the Moon (1); Lit portion changes (1).	1+1+1+1+1
	OR: Planet: large, round, clears orbit (1); Dwarf planet: round, not clear (1); Asteroid: small rocky (1); Examples: Mars, Pluto, Ceres (1); All orbit the Sun (1).	1+1+1+1+1
20	Rotation: day and night (1); Revolution: year (1); Rotation takes about 24 hours (1); Revolution takes about 365 days (1); The Sun appears to rise and set (1).	1+1+1+1+1
	OR: Very large distances (1); Light year is the distance light travels in a year (1); Sun's light takes 8 minutes (1); Stars are light years away (1); Use of such units (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Earth revolves around the Sun	Different night views.
2	**(c)** full moon	Fully lit.
3	**(b)** Earth's gravity and its speed	Balance.
4	**(d)** it is too small and does not clear its path	Dwarf planet.
5	**(c)** planet	Planets don't twinkle much.
6	**(b)** cover a large area	Wide coverage.
7	**(a)** Venus	Bright planet.
8	**(b)** Proxima Centauri	It is about four light-years away.
9	**(b)** Sriharikota	Satish Dhawan Space Centre.
10	**(d)** no air and extreme temperatures	Protection.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Meteor burns in air (1); Meteorite lands on Earth (1).	1+1
12	Very hot (1); Thick, acidic atmosphere (1).	1+1
13	Tides and light at night (1); Calendars and exploration (1).	1+1
14	No air resistance (1); Stable orbit (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It stays in the north (1); Other stars appear to turn (1); Gives directions (1).	1+1+1
16	Gravity of the Sun (1); Planets orbit (1); They do not move off (1).	1+1+1
17	Leftover rocky material (1); Could not form a planet (1); Orbit the Sun (1).	1+1+1
	OR: Inner: Mercury to Mars, rocky (1); Outer: Jupiter to Neptune, large (1); Asteroid belt in between (1).	1+1+1
18	Rotation: spin about axis, 24 hours (1); Revolution: orbit the Sun, 365 days (1); Day and year result (1).	1+1+1
	OR: GPS and maps (1); TV and phone (1); Weather forecast (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Communication: INSAT (1); Remote sensing: crops, forests (1); Navigation: NavIC (1); Weather (1); Disasters (1).	1+1+1+1+1
	OR: Float in free fall (1); Recycled air and water (1); Special food (1); Exercise (1); Experiments (1).	1+1+1+1+1
20	Craters (1); No air (1); Less gravity (1); Distance about 3.8 lakh km (1); Chandrayaan (1).	1+1+1+1+1
	OR: Sun: huge, gives light (1); Earth: medium, reflects (1); Moon: small, reflects (1); Sun gives heat (1); Others receive (1).	1+1+1+1+1

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